

Annex 9 — Equipment Configuration and Sorting Requirements

Purpose: Product catalogue of all Urban Rig models, additional pre-processing and post-processing equipment available, and a configuration matrix mapping each deployment sector to its required equipment package and associated CAPEX implications. This annex supports the sector-specific scenario economics in Annexes 10-13.

Date: March 2026

9.1 Urban Rig Product Catalogue

9.1.1 Full Product Range

Urban Rig offers seven models spanning batch, continuous, and mobile configurations (Urban Rig Recycling Solutions ENG, pp. 4-5):

Model	Type	Capacity	Zones	Target Application
URC-2000	Continuous	200 T/day	7-stage	Municipal and industrial scale
URC-1000	Continuous	100 T/day	4-stage	Mid-scale municipal
URC-400	Continuous	40 T/day	—	Small municipal / industrial
URC-150	Continuous	15 T/day	7-stage	Small-scale / remote
URB-50	Batch	5 T/batch	—	Pilot / demonstration
URB-10	Batch	1 T/batch	—	Laboratory / R&D
URT-10	Truck-mounted	1 T/batch	—	Mobile / emergency deployment

Source: Urban Rig Recycling Solutions ENG, pp. 4-5; OneWorld Corporation product documentation.

9.1.2 URC-2000 — Reference Model

The URC-2000 is the primary model for commercial deployment and the basis for all financial projections in this business plan.

Parameter	Value	Source
Volumetric capacity	200 T/day	UR2000-JP, p. 4
Effective throughput (at 0.7 T/m ³)	140 T/day	Lufthansa worksheet
Annual throughput (330 operating days)	46,200 T/year	Financial model
Dimensions	W: 66 m × D: 7 m × H: 7 m	Urban Rig Recycling Solutions ENG, pp. 6-7
Building requirement	4,650 m ² total (3,000 m ² processing)	Urban Rig Recycling Solutions ENG, p. 7
Oil storage	5 T × 8 tanks	Urban Rig Recycling Solutions ENG, p. 7
Reactor temperature	MAX 600°C normal, 0.2 MPa	Urban Rig Recycling Solutions ENG, p. 7

Parameter	Value	Source
Fuel consumption	10,000 L/day light oil	UR2000-JP, p. 4; Lufthansa worksheet
Electricity	600 kW continuous	UR2000-JP, p. 4
Personnel	~15 FTE	Urban Rig Recycling Solutions ENG, p. 19
Manufacturing lead time	~10 months	Urban Rig Recycling Solutions ENG, p. 19
All-electric option	56,800 kWh/day (replaces oil heating)	Fuel/electricity specification sheet
Base equipment price	€50.0M	OneWorld Corporation pricing

Scalability: Each URC-2000 is a self-contained, factory-manufactured module. Multiple units can be co-located: 5 × URC-2000 yields 1,000 T/day capacity on a 28,900 m² site, sharing building infrastructure and reducing per-unit CAPEX (Urban Rig Recycling Solutions ENG, p. 19).

9.1.3 URC-1000 — Mid-Scale Model

Parameter	Value	Source
Capacity	100 T/day	Urban Rig Recycling Solutions ENG, p. 5
Zones	4-stage continuous	Urban Rig Recycling Solutions ENG, p. 5
Fuel consumption	5,500 L/day	Fuel/electricity specification sheet
Electricity	350 kW continuous	Fuel/electricity specification sheet
Estimated price	~€30M per unit	Derived from Hardin LOI: USD 60M for 2 × URC-1000 = USD 30M/unit (Technical Findings Log §16.1)
Current status	Two units under construction	OneWorld Corporation

Note: The URC-1000 is relevant for the Korea Phase 1 pilot (see Annex 4 §4B.10) and for markets where 100 T/day is a better fit than 200 T/day.

9.1.4 Smaller and Mobile Models

Model	Capacity	Fuel	Electricity	Lead Time	Application
URC-400	40 T/day	2,200 L/day	250 kW	~10 months	Small municipal or industrial
URC-150	15 T/day	800 L/batch	150 kW	~10 months	Remote or island deployments
URB-50	5 T/batch	220 L/batch	50 kW	~6 months	Pilot / demonstration
URB-10	1 T/batch	120 L/batch	25 kW	~6 months	Laboratory

Model	Capacity	Fuel	Electricity	Lead Time	Application
URT-10	1 T/batch (truck)	120 L/batch	25 kW	~6 months	Emergency / disaster response

Source: Fuel/electricity specification sheet; Urban Rig Recycling Solutions ENG, pp. 4-5, 19.

Pricing for smaller models is not publicly confirmed. Scaling logic suggests approximately €8-10M per URC-400 and €3-5M per URC-150, but these are estimates only and should be verified with OneWorld Corporation.

9.2 Core System Add-Ons

These systems are designed by or for Urban Rig and can be integrated with any continuous model.

9.2.1 PRIMO Exhaust Gas Purification (PUREM 30.8)

Function: DeNO_x (ammonia mist injection), DeSO_x, dust removal, and deodorisation — using the same technology standard as thermal power plants (Urban Rig Recycling Solutions ENG, p. 12).

Parameter	Detail
Applicable models	URC-2000, URC-1000, URC-400, URC-150 (Type 6)
Purpose	Achieve near-zero atmospheric impact in sensitive environments (urban, airport)
When required	Deployments near residential areas; ICPE permitting with enhanced emission targets
Estimated CAPEX	€500k-1.5M (to be confirmed with OneWorld)

Note: Base UR emissions are already 5-50× below French incineration limits (KANSO Technos — see Annex 2 §2.7). The PRIMO system is optional for sites requiring an additional margin of safety or where permitting conditions demand best-available-technology (BAT) for air quality.

9.2.2 SEA CORAL Water Purification

Function: Ceramic purification system using sea coral as catalyst and absorbent with vaporisation heat. Purifies process water, extracting pure water free of minerals including arsenic. Enables closed-loop water management — zero wastewater discharge (Urban Rig Recycling Solutions ENG, p. 13).

Parameter	Detail
Applicable models	All continuous models
Purpose	Closed-loop water cycle, zero liquid discharge

Parameter	Detail
When required	All deployments (integrated water management for ICPE compliance)
Estimated CAPEX	Included in base package for URC-2000 (€50M) — confirm with OneWorld

9.2.3 FORCE 9 (FIA CR-09) Char Stabilisation

Function: Far-infrared catalytic reduction for char stabilisation. Converts heavy metals in char to stable, non-water-dissolved sulphur compounds at ~400°C. Patented technology (Urban Rig Recycling Solutions ENG, pp. 10–11).

Parameter	Detail
Purpose	Ensures char output is environmentally safe for soil application or sale as fuel
When required	All deployments where char is sold as biochar (Tier 4–5 markets — see Annex 3 §3.2.2)
Estimated CAPEX	Included in base package — confirm with OneWorld

9.3 Pre-Processing Equipment

Pre-processing equipment is required between waste reception and reactor input to remove non-processable materials and size-reduce the feedstock. Requirements vary by sector.

9.3.1 Mechanical Pre-Sorting Line

Component	Function	When Required
Trommel screen	Size classification, removal of oversized items	All MSW and mixed-waste deployments
Magnetic separator	Ferrous metal extraction (steel cans, nails, wire)	All deployments
Eddy-current separator	Non-ferrous metal extraction (aluminium, copper)	All deployments
Optical sorter (NIR)	Identification and removal of glass, specific contaminants	Optional — for high-contamination feedstocks
Ballistic separator	Separation of 2D (films) from 3D (containers)	Optional — for mixed packaging streams
Manual sorting station	Quality control, hazardous item removal	Recommended for all deployments

Estimated CAPEX for complete pre-sorting line: €2–5M depending on throughput capacity and automation level. This is included in the “Building and site preparation” line of the financial model (€5M for France deployment).

9.3.2 Size Reduction

Component	Function	When Required
Primary shredder	Reduce waste to <300 mm	All deployments
Secondary shredder/granulator	Further reduction to <50 mm for reactor input	May be required for bulky or rigid waste streams

Note: The URC-2000 reactor bucket is 1.5 m × 3 m × 1 m (4.5 m³). Size reduction to <300 mm is sufficient for most feedstocks. The Urban Rig Recycling Solutions ENG brochure states the system accepts waste “without requiring pre-sorting or cleaning beyond basic size reduction and inert removal” (p. 6).

9.4 Post-Processing Equipment

9.4.1 Oil Storage and Handling

The base URC-2000 package includes 8 × 5T oil storage tanks (Urban Rig Recycling Solutions ENG, p. 7). Additional storage may be required for:

- **Separate fraction storage:** 3 tank banks (light oil, gas oil, heavy oil) to preserve price differentiation (see Annex 3 §3.1.6)
- **Tanker loading facility:** For oil dispatch to buyers

Estimated additional CAPEX: €200k–500k for expanded tank farm and loading.

9.4.2 Char Pelletisation

Required if char is sold as industrial fuel pellets (Tier 2) or charcoal briquettes (Tier 3) rather than loose material:

Parameter	Detail
Purpose	Compress char into pellets/briquettes for handling and sale
Throughput required	~67 T/day (all combustible char output)
Estimated CAPEX	€200k–500k per unit

Source: Annex 5 §13.6.

9.4.3 Char Combustion System

No longer required in V2 model: The updated V2 configuration uses purchased heating oil for all reactor heating. All char output is sold, eliminating the need for a char combustion system. This removes the associated CAPEX of €500k–1M.

9.5 Sector-Specific Configuration Matrix

The following matrix maps each deployment sector (Annexes 5–8) to its required equipment package. “Base” means included in the €50M URC-2000 package. “Add” means additional CAPEX.

Equipment	MSW (Annex 5)	CDG Airport (Annex 6)	Industrial (Annex 7)	Plastic Rejects (Annex 8)
Core reactor (URC-2000)	Base	Base	Base	Base
PRIMO exhaust (PUREM)	Optional	Recommended	Optional	Optional
SEA CORAL water	Base	Base	Base	Base
FORCE 9 char stabilisation	Base	Base	Base	Base
Trommel screen	Add	Add	Add	Not needed
Magnetic separator	Add	Add	Add	Add (minimal)
Eddy-current separator	Add	Add	Add	Not needed
Optical sorter (NIR)	Optional	Optional	Optional	Not needed
Primary shredder	Add	Add	Add	Add
Secondary shredder	Optional	Optional	Optional	Optional
Manual sorting station	Add	Add	Add	Optional
Expanded oil tank farm	Add	Add	Add	Add
Char pelletiser	Add	Add	Add	Not needed

9.5.1 Estimated CAPEX by Sector

CAPEX Component	MSW	CDG Airport	Industrial	Plastic Rejects
URC-2000 base package	€50.0M	€50.0M	€50.0M	€50.0M
Building and site preparation	€5.0M	€5.0M	€5.0M	€3.0M
Pre-sorting line	€3.0M	€3.5M	€2.5M	€1.0M
PRIMO (if selected)	—	€1.0M	—	—
Char pelletiser	€0.3M	€0.3M	€0.3M	—
Oil storage expansion	€0.3M	€0.3M	€0.3M	€0.3M
Installation, freight, training	€0.9M	€0.9M	€0.9M	€0.9M
Accessories, spares, monitoring	€2.0M	€2.0M	€2.0M	€2.0M
Contingency (5%)	€2.5M	€2.6M	€2.5M	€2.9M
TOTAL CAPEX	€64.6M	€66.2M	€64.1M	€60.1M

Notes:

- The current financial model uses €60.8M CAPEX (UR-Financial-Model-General.xlsx, CAPEX sheet). The sector-specific estimates above show that MSW and CDG deployments may require €65–67M when pre-sorting, char combustion, and PRIMO are included. This should be reflected in the scenario economics (Annexes 10–13).
- Plastic sorting rejects require minimal pre-processing (material is already sorted and baled) — CAPEX is correspondingly lower.
- Building costs for Korea are €6M (higher land prices — see Annex 4 §4B).

9.6 Recommendations

1. **Confirm pricing for PRIMO, SEA CORAL, and FORCE 9** with OneWorld Corporation — specifically whether these are included in the €50M base package or are additional items.
 2. **Obtain firm quotes for pre-sorting equipment** from European MRF equipment suppliers (TOMRA, Pellenc ST, Machinex) for integration with the site design.
 3. **Request URC-1000 and URC-400 pricing** from OneWorld for Korea Phase 1 and smaller-market deployments.
 4. **Consider pre-sorting as a CAPEX line item in all sector models** — the current financial model bundles it into “Building and site preparation” (€5M), which may be insufficient for MSW and airport deployments requiring a full MRF-style line.
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9.7 References

Urban Rig Technical Sources

1. **Urban Rig Recycling Solutions ENG** — OneWorld Corporation product brochure (English edition). Cited for: full product range (pp. 4–5), URC-2000 dimensions and specifications (pp. 6–7), oil storage (p. 7), reactor temperature and pressure (p. 7), PRIMO exhaust system (p. 12), SEA CORAL water purification (p. 13), FORCE 9 char stabilisation (pp. 10–11), automation (p. 17), personnel and lead time (p. 19), multi-unit scalability (p. 19).
2. **UR2000-JP** — URC-2000 technical specification sheet (Japanese edition). Cited for: volumetric capacity 200 T/day, fuel consumption, electricity (p. 4).
3. **Fuel/electricity specification sheet** — OneWorld Corporation. Cited for: fuel and electricity consumption across all models, all-electric option 56,800 kWh/day (§9.1.2, §9.1.4).
4. **OneWorld Corporation** — Product pricing and documentation. Cited for: URC-2000 base price €50M, URC-1000 current status (§9.1.2, §9.1.3).

Financial and Operational Sources

5. **Lufthansa worksheet** — Operational assumptions. Cited for: effective throughput at 0.7 T/m³ density (§9.1.2).
6. **Hardin LOI** — Letter of Intent for 2 × URC-1000 at USD 60M. Via Technical Findings Log §16.1. Cited for: URC-1000 estimated price ~€30M/unit (§9.1.3).
7. **UR-Financial-Model-General.xlsx** — Working financial model (CAPEX sheet). Cited for: current model CAPEX of €60.8M (§9.5.1, Notes).

Cross-Referenced Annexes

8. **Annex 2 §2.6.3** — Energy balance derivation. Cited for: V2 char sales model with purchased heating oil supply (§9.4.3).
 9. **Annex 2 §2.7** — KANSO Technos emission data. Cited for: base emissions 5–50× below French incineration limits (§9.2.1).
 10. **Annex 3 §3.1.6** — Oil fraction storage recommendation. Cited for: separate tank banks (§9.4.1).
 11. **Annex 3 §3.2.2** — Biochar market tiers. Cited for: char as fuel or biochar (§9.2.3).
 12. **Annex 4 §4B** — Korea country study. Cited for: Korea building costs €6M, Phase 1 pilot with URC-1000 (§9.5.1, Notes).
 13. **Annex 5 §13.6** — Char pelletisation. Cited for: pelletiser CAPEX reference (§9.4.2).
 14. **Annexes 5–8** — Sector-specific feedstock profiles. Cited for: sector configuration requirements (§9.5).
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End of Annex 9